

Tech Corp Analytics

SQL Business Intelligence & Business Analytics Project

Prepared By

Sandesh Navnath Nangare

Aspiring Data Analyst

Tools Used

- MySQL
- Microsoft Excel
- Microsoft Word
- Power BI (*Upcoming Dashboard Development*)
- GitHub

Skills Demonstrated

- SQL Querying
- Joins (INNER JOIN, LEFT JOIN)
- Aggregate Functions
- Window Functions
- Common Table Expressions (CTEs)
- Ranking Functions
- Business Analysis
- Data Interpretation
- Business Reporting
- Analytical Thinking

Tech Corp Analytics Project

Project Introduction

- Analyse employees, departments, locations, clients, and projects to help management understand workforce distribution, project allocation, and business performance.

Step 1 :- Data Understanding

1. Employee Table

Contains:

- employee_id
- employee_name
- department
- location
- salary
- experience

Purpose:

- Workforce analysis

2. Department Table

Contains:

- department_id
- department_name

Purpose:

- Department-level reporting

3. Project Table

Contains:

- project_id
- project_name

- client_id
- department_id
- project_status
- project_budget

Purpose:

- Project and budget analysis

4. Client Table

Contains:

- client_id
- client_name
- industry

Purpose:

- Client and industry-level business analysis

5. Location Table

Contains:

- location_id
- city

Purpose:

- Location-wise workforce and salary analysis

Business Questions

Employee Analysis

- Business Question 1

How many employees are in each department?

- Business Objective

Understand workforce distribution across departments and identify departments with high or low staffing levels.

- SQL Query

```
WITH emp_dept AS (  
  
SELECT d.department_name, COUNT(*) AS total_employee  
  
FROM employee e JOIN department d  
  
ON e.department_id = d.department_id  
  
GROUP BY d.department_name )  
  
SELECT department_name, total_employee,  
  
ROUND(total_employee * 100.0 / SUM(total_employee) OVER (),2 ) AS percentage  
  
FROM emp_dept;
```

- Query Result

Department Name	Total Employee	Percentage
Data Analytics	526	10.52%
Finance	492	9.84%
HR	488	9.76%
Marketing	478	9.56%
Sales	501	10.02%
IT	446	8.92%
Cyber Security	524	10.48%
Operations	512	10.24%
Product	520	10.4%
Customer Success	513	10.26%

- Business Insight

The IT department represents the largest share of the workforce, indicating a strong operational focus on technology functions.

- ## Business Question 2

How many employees are located in each city?

- ## Business Objective

Analyze workforce presence across locations.

- ## SQL Query

```
SELECT
    l.city, COUNT(*) AS total_emp_per_city
FROM
    employee e
    JOIN
    location l ON e.location_id = l.location_id
GROUP BY l.city;
```

- ## Query Result

City	Total Emp Per City
Mumbai	507
Pune	460
Bangalore	494
Hyderabad	492
Chennai	510
Delhi	535
Noida	476
Gurgaon	509
Kolkata	506
Ahmedabad	511

- ## Business Insight

Cities with high employee concentration may be strategic operational hubs.

- ## Business Question 3

What is the gender distribution across the company?

- ## Business Objective

Assess workforce diversity.

- ## SQL Query

```
SELECT
    gender,
    COUNT(*) AS employee_count,
    ROUND(
        COUNT(*) * 100.0 /
        SUM(COUNT(*)) OVER (),
        2
    ) AS percentage_of_workforce
FROM employee
GROUP BY gender;
```

- ## Query Result

Gender	Employee count	Percentage of Workforce
Male	2466	49.32 %
Female	2534	50.68 %

- ## Business Insight

Large imbalances may indicate opportunities to improve diversity

- ## Business Question 4

What is the average salary by department?

- ## Business Objective

Compare compensation across departments.

- SQL Query

```
SELECT
    d.department_name, AVG(e.salary) AS avg_salary
FROM
    department d
    LEFT JOIN
        employee e ON e.department_id = d.department_id
GROUP BY d.department_name
ORDER BY avg_salary DESC;
```

- Query Result

Department Name	AVG salary
Sales	142346.0299
HR	140205.416
Product	139183.8558
Customer Success	138143.5984
Data Analytics	137705.6635
IT	137369.5045
Marketing	137076.8473
Cyber Security	136939.9485
Finance	136400.8923
Operations	135156.6191

- Business Insight

Higher average salaries may reflect specialized skill requirements.

- Business Question 5

What is the average experience by department?

- Business Objective

Measure workforce seniority by department.

- SQL Query

```
SELECT
    d.department_name,
    ROUND(AVG(e.experience_years), 2) AS avg_experience
```

```
FROM department d

LEFT JOIN employee e

    ON d.department_id = e.department_id

GROUP BY

    d.department_id,

    d.department_name

ORDER BY avg_experience DESC;
```

- Query Result

Department Name	AVG Experience
Customer Success	10.18 %
Data Analytics	10.14 %
Cyber Security	10.13 %
Finance	10.09 %
Marketing	10.06 %
Operations	10.05 %
IT	10.05 %
Product	10.01 %
HR	9.86 %
Sales	9.78 %

- Business Insight

Departments with lower experience levels may need mentoring or training.

- Business Question 6

Which employees earn more than their department's average salary?

- Business Objective

Identify employees compensated above peer averages.

- SQL Query

```
WITH avg_salary_department AS

(SELECT

    d.department_id ,d.department_name, AVG(e.salary) AS dept_avg

FROM

    department d

    LEFT JOIN
```



```

        employee e ON e.department_id = d.department_id

GROUP BY d.department_id ,d.department_name)

SELECT

    e.employee_name,

    e.salary,

    dept_salary.department_name,

    dept_salary.dept_avg,

    (e.salary - dept_salary.dept_avg) AS salary_difference

FROM

    avg_salary_department dept_salary

    JOIN

    employee e ON dept_salary.department_id = e.department_id

WHERE

    e.salary > dept_salary.dept_avg

ORDER BY salary_difference DESC;

```

• Query Result

Employee name	Salary	Department Name	Dept AVG	Salary Difference
Employee_3678	249698	Operations	135156.6	114541.3809
Employee_288	248840	Operations	135156.6	113683.3809
Employee_3305	248830	Operations	135156.6	113673.3809
Employee_2486	249935	Finance	136400.9	113534.1077
Employee_1870	248481	Operations	135156.6	113324.3809
Employee_4361	249821	Cyber Security	136939.9	112881.0515
Employee_3814	249281	Finance	136400.9	112880.1077
Employee_2036	249748	Cyber Security	136939.9	112808.0515
Employee_2294	247958	Operations	135156.6	112801.3809
Employee_963	249716	Cyber Security	136939.9	112776.0515

Total employees returned: 2,509

• Business Insight

May highlight senior specialists or compensation outliers.

- Business Question 7

Which employees earn more than their department's average salary and have above-average experience?

- Business Objective

Identify highly experienced and highly compensated employees.

- SQL Query

```
WITH department_stats AS

(SELECT

    d.department_id,

    d.department_name,

    AVG(e.salary) AS avg_salary,

    AVG(experience_years) AS avg_exp

FROM

    employee e

    JOIN

    department d ON e.department_id = d.department_id

GROUP BY d.department_id,d.department_name)

SELECT

    e.employee_name,

    t.department_name,

    e.salary,

    e.experience_years

FROM

    employee e

    JOIN

    department_stats t ON e.department_id = t.department_id

WHERE

    e.salary > avg_salary

    AND e.experience_years > avg_exp;
```

- Query Result

Employee Name	Department Name	Salary	Experience Years
Employee_1	Customer Success	231633	15
Employee_2	Finance	234466	19
Employee_7	Operations	214939	12
Employee_8	HR	235736	16
Employee_9	Product	231697	16
Employee_10	Sales	182143	17
Employee_13	Marketing	176147	20
Employee_16	Sales	233777	15
Employee_17	Finance	236322	14
Employee_18	Sales	222532	15

Total employees returned: 1,217

- Business Insight

These employees may be key contributors or future leaders

- Business Question 8

Show the Top 3 highest-paid employees within each department.

- Business Objective

Identify top earners within each department.

- SQL Query

```
WITH employee_salary_rank AS
( SELECT
    e.employee_name,
    d.department_name,
    e.salary,
    ROW_NUMBER() OVER (
        PARTITION BY e.department_id
        ORDER BY e.salary DESC) AS salary_rank
FROM employee e
JOIN department d
    ON e.department_id = d.department_id)
SELECT *
FROM employee_salary_rank
```

WHERE salary_rank <= 3

ORDER BY

department_name,

salary_rank;

- Query Result

Employee Name	Department name	Salary	Salary Rank
Employee_220	Customer Success	249670	1
Employee_1078	Customer Success	248677	2
Employee_1504	Customer Success	248443	3
Employee_4361	Cyber Security	249821	1
Employee_2036	Cyber Security	249748	2
Employee_963	Cyber Security	249716	3
Employee_2533	Data Analytics	248791	1
Employee_1815	Data Analytics	248742	2
Employee_2399	Data Analytics	248621	3
Employee_2486	Finance	249935	1
Employee_3814	Finance	249281	2

Total employees returned: 30

- Business Insight

Useful for compensation benchmarking and leadership reviews.

Department Analysis

- Business Question 1

Which department has the largest workforce?

- Business Objective

Identify the biggest department.

- SQL Query

```
SELECT
    d.department_name, COUNT(*) AS department_workforce
FROM
    employee e
    JOIN
    department d ON e.department_id = d.department_id
GROUP BY d.department_name
ORDER BY department_workforce DESC;
```

- Query Result

Department Name	Department Workforce
Data Analytics	526
Cyber Security	524
Product	520
Customer Success	513
Operations	512
Sales	501
Finance	492
HR	488
Marketing	478
IT	446

- Business Insight

Large departments often have the greatest operational impact.

- **Business Question 2**

Which department has the highest average salary and how many employee in that department?

- **Business Objective**

Compare department compensation levels.

- **SQL Query**

```
SELECT  
  
    COUNT(*) AS employee_count,  
  
    d.Department_Name,  
  
    ROUND(AVG(e.salary), 2) AS Department_Avg_Salary  
  
FROM  
  
    department d  
  
    LEFT JOIN  
  
    employee e ON e.department_id = d.department_id  
  
GROUP BY d.department_name , d.department_id  
  
ORDER BY Department_avg_salary DESC  
  
LIMIT 1;
```

- **Query Result**

Employee Count	Department Name	Department AVG Salary
501	Sales	142346.03

- **Business Insight**

The sale department has the highest average salary with 501 employees, indicating that it may require highly specialized skills or senior-level expertise.

- **Business Question 3**

Which department has the highest total salary expense?

- **Business Objective**

Understand salary cost allocation.

- **SQL Query**

```
SELECT
    d.Department_Name,
    COUNT(e.employee_id) AS Employee_Count,
    SUM(e.salary) AS Total_Salary_Expense
FROM department d
LEFT JOIN employee e
    ON d.department_id = e.department_id
GROUP BY
    d.department_id,
    d.department_name
ORDER BY total_salary_expense DESC
LIMIT 1;
```

- **Query Result**

Department Name	Employee Count	Total Salary Expense
Data Analytics	526	72433179

- **Business Insight**

High-cost departments should be evaluated against business value.

- **Business Question 4**

Which department has the most experienced workforce on average?

- **Business Objective**

Measure department expertise.

- **SQL Query**

```
SELECT
```

```

d.Department_Name,
ROUND(AVG(e.experience_years),2) AS Avg_Experience_Years,
COUNT(e.employee_id) AS Total_Emp_Per_Dept
FROM
employee e
LEFT JOIN
department d ON e.department_id = d.department_id
GROUP BY d.department_id ,d.department_name
ORDER BY avg_experience_years DESC limit 1;

```

- **Query Result**

Department Name	Avg Experience Years	Total Emp Per Dept
Customer Success	10.18	513

- **Business Insight**

Experienced teams may be better positioned for complex work.

- **Business Question 5**

How many projects are managed by each department?

- **Business Objective**

Measure the number of projects handled by each department to evaluate workload distribution, support resource planning, and identify departments with the highest operational responsibility.

- **SQL Query**

```

SELECT
d.department_name, COUNT(p.project_id) AS count_of_projects
FROM
department d
LEFT JOIN
project p ON p.department_id = d.department_id

```


GROUP BY d.department_id , d.department_name

ORDER BY count_of_projects DESC;

- Query Result

Department Name	Count of Projects
IT	107
Customer Success	107
Operations	105
HR	103
Data Analytics	99
Marketing	99
Sales	98
Cyber Security	98
Product	93
Finance	91

- Business Insight

The IT and Customer Success department manages the highest number of projects, indicating a greater operational workload and a significant contribution to the organization's project delivery.

- Business Question 6

What is the total project budget managed by each department?

- Business Objective

Calculate the total project budget managed by each department to evaluate budget allocation, identify high-investment departments, and support financial planning and resource allocation.

- SQL Query

SELECT

d.Department_Name, COALESCE(SUM(p.project_budget), 0) AS
total_budget

FROM

department d

LEFT JOIN

```
project p ON d.department_id = p.department_id  
GROUP BY d.department_id , d.department_name  
ORDER BY total_budget DESC;
```

- Query Result

Department Name	Total Budget
IT	292128969
Customer Success	285321218
HR	278445625
Operations	270671053
Cyber Security	264872852
Product	256170139
Data Analytics	251176807
Marketing	246312350
Finance	243456069
Sales	223109210

- Business Insight

The IT department manages the highest total project budget, indicating it oversees high-value projects and plays a significant role in the organization's strategic initiatives.

- Business Question 7

What is the average project budget managed by each department?

- Business Objective

Calculate the average project budget for each department to compare the typical investment per project and identify departments managing higher-value initiatives.

- SQL Query

```
SELECT  
  
    d.Department_Name, COALESCE(ROUND(avg(p.project_budget),2), 0)  
    AS AVG_Project_budget  
  
FROM  
  
    department d
```

LEFT JOIN

project p ON d.department_id = p.department_id

GROUP BY d.department_id , d.department_name

ORDER BY AVG_Project_budget DESC;

- Query Result

Department Name	AVG Project budget
Product	2754517.62
IT	2730177.28
HR	2703355.58
Cyber Security	2702784.2
Finance	2675341.42
Customer Success	2666553.44
Operations	2577819.55
Data Analytics	2537139.46
Marketing	2488003.54
Sales	2276624.59

- Business Insight

The Product department has the highest average project budget, indicating that it typically handles projects requiring greater financial investment and strategic importance.

- Business Question 8

Which departments have completed the highest number of projects?

- Business Objective

Measure the number of completed projects handled by each department to evaluate project delivery performance and identify departments with the strongest execution record.

- SQL Query

SELECT

d.department_name,

SUM(

CASE

```

        WHEN p.project_status = 'Completed' THEN 1
        ELSE 0
    END
) AS completed_projects_count
FROM department d
LEFT JOIN project p
    ON d.department_id = p.department_id
GROUP BY
    d.department_id,
    d.department_name
ORDER BY
    completed_projects_count DESC;

```

- Query Result

Department Name	Completed Projects Count
IT	34
Operations	30
Cyber Security	29
Data Analytics	25
HR	25
Customer Success	24
Marketing	23
Sales	22
Product	22
Finance	20

- Business Insight

The IT department has completed the highest number of projects, indicating strong project execution and successful delivery performance.

- Business Question 9

Which departments have the highest number of ongoing projects?

- **Business Objective**

Analyze the number of projects currently in progress for each department to monitor active workloads, support resource allocation, and identify departments with the highest ongoing operational responsibilities.

- **SQL Query**

```
SELECT
    d.department_name,
    SUM(CASE
        WHEN p.project_status = 'in progress' THEN 1
        ELSE 0
    END) AS in_progress_count
FROM
    department d
    LEFT JOIN
        project p ON d.department_id = p.department_id
GROUP BY d.department_id , d.department_name
ORDER BY in_progress_count DESC;
```

- **Query Result**

Department Name	In Progress Count
Customer Success	27
Finance	26
Sales	24
Cyber Security	24
Product	23
Data Analytics	21
IT	21
HR	20
Marketing	20
Operations	18

- **Business Insight**

The Customer Success department is managing the highest number of projects currently in progress, indicating a significant active workload and a greater demand for operational resources.

- **Business Question 10**

What percentage of the company's total project budget is managed by each department?

- **Business Objective**

Calculate each department's percentage share of the company's total project budget to understand budget distribution, identify departments managing the largest investments, and support strategic financial planning.

- **SQL Query**

```
WITH department_budgets AS
(SELECT
    d.department_id,
    d.department_name,
    SUM(p.project_budget) AS department_total_project_budget,
    (SELECT
        SUM(p.project_budget)
    FROM
        project p) AS company_budget
FROM
    department d
    LEFT JOIN
        project p ON d.department_id = p.department_id
GROUP BY d.department_id , d.department_name)
```

```

SELECT
    department_id,
    department_name,
    department_total_project_budget,
    company_budget,
    ROUND(((db.department_total_project_budget / db.company_budget) *
100),
        2)
        AS percentage_of_project_budgets
FROM
    department_budgets db
ORDER BY percentage_of_project_budgets DESC;

```

- Query Result

Department Id	Department Name	Total Project Budget	Company Budget	Percentage of Project Budgets
6	IT	292128969	2611664292	11.19
10	Customer Success	285321218	2611664292	10.92
3	HR	278445625	2611664292	10.66
8	Operations	270671053	2611664292	10.36
7	Cyber Security	264872852	2611664292	10.14
9	Product	256170139	2611664292	9.81
1	Data Analytics	251176807	2611664292	9.62
4	Marketing	246312350	2611664292	9.43
2	Finance	243456069	2611664292	9.32
5	Sales	223109210	2611664292	8.54

- Business Insight

The IT department manages the largest share of the company's total project budget, indicating that it oversees a significant portion of organizational investment and strategic projects.

Project Analysis

- Business Question 1

Which projects have the highest budgets?

- Business Objective

Identify the projects with the highest budgets to understand where the company is making its largest financial investments and support strategic budget monitoring.

- SQL Query

WITH budgets AS

```
(  
    SELECT  
        project_id,  
        project_name,  
        project_status,  
        project_budget,  
        RANK() OVER(ORDER BY project_budget DESC) AS project_rank  
    FROM project  
)  
SELECT  
    project_id,  
    project_name,  
    project_status,  
    project_budget,  
    project_rank  
FROM  
    budgets  
WHERE  
    project_rank <= 10
```


ORDER BY project_rank;

- Query Result

Project Id	Project Name	Project Status	Project Budget	Project Rank
2	Project_2	Completed	4993777	1
383	Project_383	Planning	4993777	1
656	Project_656	Completed	4985425	3
731	Project_731	Planning	4978175	4
741	Project_741	Completed	4974356	5
387	Project_387	Planning	4971743	6
311	Project_311	Completed	4958009	7
849	Project_849	In Progress	4956343	8
168	Project_168	Completed	4951856	9
3	Project_3	Planning	4950054	10
832	Project_832	Completed	4950054	10

- Business Insight

The top-ranked projects account for the company's largest financial commitments. These projects require close monitoring to ensure budgets are effectively managed and strategic objectives are achieved.

- Business Question 2

What is the distribution of projects by status?

- Business Objective

Analyze the distribution of projects across different statuses to evaluate the company's project pipeline, monitor operational progress, and support project management decisions.

- SQL Query

```
SELECT
    project_status, COUNT(project_status) AS total_projects
FROM
    project
GROUP BY project_status
ORDER BY total_projects DESC;
```

- Query Result

Project Status	Total Projects
Planning	261
On Hold	261
Completed	254
In Progress	224

- Business Insight

The project status with the highest count represents the largest portion of the company's project pipeline, helping management understand where most projects are currently concentrated.

- Business Question 3

What is the average project budget by project status?

- Business Objective

Calculate the average project budget for each project status to compare investment levels across completed, in-progress, planning, and on-hold projects, supporting budget evaluation and project planning.

- SQL Query

```
SELECT
    project_status,
    ROUND(AVG(project_budget), 2) AS avg_project_budgets
FROM
    project
GROUP BY project_status
ORDER BY avg_project_budgets DESC;
```

- Query Result

Project Status	AVG Project Budgets
Planning	2653793.7
On Hold	2640111.47
Completed	2580560.97
In Progress	2564698.91

- **Business Insight**

The project status with the highest average budget represents the category receiving the greatest investment per project, indicating where the organization is allocating larger financial resources.

- **Business Question 4**

Which are the Top 10 highest-budget projects and which departments manage them?

- **Business Objective**

Identify the Top 10 highest-budget projects and the departments responsible for managing them to highlight the company's largest financial investments and support executive budget monitoring and strategic decision-making.

- **SQL Query**

WITH high AS

```
(  SELECT
    RANK() OVER(ORDER BY p.project_budget DESC) AS ranks,
    p.project_name,
    d.department_name,
    p.project_budget
  FROM project p
  LEFT JOIN department d
    ON p.department_id = d.department_id)
```

```
SELECT
  ranks,
  project_name,
  department_name,
```

```
project_budget
FROM high
WHERE ranks <= 10;
```

- Query Result

Ranks	Project Name	Department Name	Project Budget
1	Project_383	Operations	4993777
2	Project_656	Product	4985425
3	Project_731	HR	4978175
4	Project_741	Data Analytics	4974356
5	Project_387	Data Analytics	4971743
6	Project_311	Data Analytics	4958009
7	Project_849	Marketing	4956343
8	Project_168	Operations	4951856
9	Project_832	Operations	4950054
10	Project_199	IT	4943704

- Business Insight

The Top 10 highest-budget projects represent the company's largest financial commitments. The departments managing these projects play a critical role in executing high-value strategic initiatives and require close performance and budget monitoring.

- Business Question 5

Which clients have the highest number of projects?

- Business Objective

Identify clients with the highest number of projects to understand client engagement, strengthen key business relationships, and prioritize account management efforts.

- SQL Query

```
SELECT
    c.client_id,
    c.client_name,
    COUNT(p.project_id) AS project_count
```

```
FROM
    client c
    LEFT JOIN
    project p ON p.client_id = c.client_id
GROUP BY c.client_id , c.client_name
ORDER BY project_count DESC;
```

- Query Result

Client ID	Client Name	Project Count
126	Client_126	9
45	Client_45	7
108	Client_108	7
247	Client_247	7
36	Client_36	6
107	Client_107	6
110	Client_110	6
116	Client_116	6
138	Client_138	6

- Business Insight

Clients with the highest number of projects represent the organization's most active business relationships. These clients contribute significantly to the company's project portfolio and may offer opportunities for long-term partnerships and additional business.

- Business Question 6

Which clients have the highest total project budget?

- Business Objective

Calculate the total project budget for each client to identify the company's highest-value clients and support strategic account management, revenue planning, and business growth initiatives.

- SQL Query

```
SELECT
    c.client_name,
```

```

        COALESCE(sum(p.project_budget),0) AS total_project_budget
FROM
    client c
    LEFT JOIN
        project p ON p.client_id = c.client_id
GROUP BY c.client_id , c.client_name
ORDER BY total_project_budget DESC;

```

- **Query Result**

Client Name	Total Project Budget
Client_108	25910167
Client_116	25526240
Client_126	21935872
Client_336	19745818
Client_187	19345805
Client_202	19176092
Client_110	17181311
Client_387	16980241
Client_290	16700472
Client_222	16600891

- **Business Insight**

Clients with the highest total project budgets contribute the greatest financial value to the organization. These clients represent strategic business relationships that should be carefully managed and nurtured.

- **Business Question 7**

Which projects are currently on hold, and which departments are responsible for them?

- **Business Objective**

Identify projects that are currently on hold along with the departments responsible for them, enabling management to monitor project delays, assess risks, and prioritize recovery efforts.

- SQL Query

```
SELECT
    p.project_name,
    d.department_name AS responsible_department,
    p.project_budget,
    p.project_status
FROM
    project p
    LEFT JOIN
    department d ON p.department_id = d.department_id
WHERE
    p.project_status = 'on hold'
ORDER BY p.project_budget DESC;
```

- Query Result

Project Name	Responsible Department	Project Budget	Project Status
Project_699	Customer Success	4939020	On Hold
Project_580	HR	4895481	On Hold
Project_232	HR	4863846	On Hold
Project_916	Customer Success	4857044	On Hold
Project_467	IT	4854443	On Hold
Project_598	Finance	4852396	On Hold
Project_275	Product	4848061	On Hold
Project_692	Product	4847157	On Hold
Project_239	Customer Success	4835093	On Hold
Project_579	Sales	4764686	On Hold

- Business Insight

On-hold projects indicate potential operational challenges such as resource constraints, budget issues, or changing business priorities. Identifying the responsible departments helps management investigate the underlying causes and take corrective action.

- **Business Question 8**

Which project statuses account for the highest total project budget?

- **Business Objective**

Calculate the total project budget for each project status to understand how the company's financial investments are distributed across different stages of the project lifecycle and support strategic budget planning.

- **SQL Query**

```
SELECT  
  
    project_status, COALESCE(SUM(project_budget),0) AS  
total_project_budget  
  
FROM  
  
    project  
  
GROUP BY project_status  
  
ORDER BY total_project_budget DESC;
```

- **Query Result**

Project Status	Total Project Budget
Planning	692640156
On Hold	689069093
Completed	655462487
In Progress	574492556

- **Business Insight**

The project status with the highest total budget represents the largest share of the company's financial investment. This helps management understand where organizational resources are currently concentrated.

Client Analysis

- **Business Question 1**

Which clients belong to each industry?

- **Business Objective**

Categorize clients by industry to understand the company's customer base and support market segmentation strategies.

- **SQL Query**

```
select client_name ,industry from client  
  
ORDER BY industry, client_name asc;
```

- **Query Result**

Client Name	Industry
Client_104	Banking
Client_107	Banking
Client_108	Banking
Client_11	Banking
Client_111	Banking
Client_12	Banking
Client_125	Banking
Client_128	Banking
Client_130	Banking
Client_134	Banking

- **Business Insight**

The report provides a clear view of the industries served by the company, helping management understand customer distribution across different market sectors.

- **Business Question 2**

Which industry has the highest number of clients?

- **Business Objective**

Identify the industry with the highest number of clients to understand the company's strongest market segment and support strategic business development initiatives.

- **SQL Query**

```
SELECT  
  
    COUNT(client_name) AS client_counts, industry  
  
FROM  
  
    client  
  
GROUP BY industry  
  
ORDER BY client_counts DESC  
  
LIMIT 1;
```

- **Query Result**

Client Counts	Industry
117	Manufacturing
102	Banking
97	Retail
95	Ecommerce
89	Healthcare

- **Business Insight**

The industry with the highest client count represents the company's largest customer segment, indicating where the business has established a strong market presence.

- **Business Question 3**

Which industry generates the highest total project budget?

- **Business Objective**

Calculate the total project budget for each client industry to identify the sectors contributing the greatest financial value and support strategic investment and business development decisions.

- **SQL Query**

```
SELECT  
  
    c.industry, COALESCE(SUM(p.project_budget),0) AS  
total_project_budget  
  
FROM  
  
    client c  
  
    LEFT JOIN  
  
    project p ON c.client_id = p.client_id  
  
GROUP BY c.industry  
  
ORDER BY total_project_budget DESC;
```

- **Query Result**

Industry	Total Project Budget
Manufacturing	607378072
Banking	574511135
Retail	549573280
Healthcare	448272204
Ecommerce	431929601

- **Business Insight**

Industries with the highest total project budgets represent the company's most valuable market segments, highlighting where the organization generates the greatest project-based revenue and investment.

- **Business Question 4**

Which clients have the highest average project budget?

- **Business Objective**

Calculate the average project budget for each client to identify customers with the highest average investment per project and support premium client relationship management.

- **SQL Query**

```
SELECT  
  
    c.client_id,c.client_name,  
  
    COALESCE(ROUND(AVG(p.project_budget), 2), 0) AS  
    avg_project_budget  
  
FROM  
  
    client c  
  
    LEFT JOIN  
  
    project p ON c.client_id = p.client_id  
  
GROUP BY c.client_id , C.client_name  
  
ORDER BY avg_project_budget DESC LIMIT 1;
```

- **Query Result**

Client ID	Client Name	AVG Project Budget
297	Client_297	4993777

- **Business Insight**

Clients with the highest average project budget represent premium engagements that may require specialized account management and strategic attention.

- **Business Question 5**

Which clients have the highest number of completed projects?

- **Business Objective**

Identify clients with the highest number of completed projects to recognize strong customer relationships, evaluate successful project delivery, and support strategic account management.

- **SQL Query**

```
SELECT
    c.client_name, COUNT(p.project_status) AS completed_projects
FROM
    client c
    JOIN
    project p ON c.client_id = p.client_id
WHERE
    p.project_status = 'completed'
GROUP BY c.client_id , c.client_name
ORDER BY completed_projects DESC;
```

- **Query Result**

Client Name	Completed Projects
Client_405	3
Client_18	3
Client_242	3
Client_187	3
Client_279	3
Client_45	3
Client_256	3
Client_247	3
Client_438	3
Client_496	2

- **Business Insight**

Clients with the highest number of completed projects represent long-term and successful business relationships, indicating strong customer engagement and consistent project execution.

Location Analysis

- Business Question 1

Which locations have the highest number of employees?

- Business Objective

Analyze employee distribution across office locations to identify workforce concentration and support resource allocation, staffing decisions, and future office expansion planning.

- SQL Query

```
SELECT  
  
    l.city, COUNT(e.employee_id) AS total_employee  
  
FROM  
  
    location l  
  
    LEFT JOIN  
  
    employee e ON l.location_id = e.location_id  
  
GROUP BY l.location_id , l.city  
  
ORDER BY total_employee DESC;
```

- Query Result

City	Total Employee
Delhi	535
Ahmedabad	511
Chennai	510
Gurgaon	509
Mumbai	507
Kolkata	506
Bangalore	494
Hyderabad	492
Noida	476
Pune	460

- **Business Insight**

Locations with the highest employee counts represent the company's largest operational hubs and may require greater investment in infrastructure, HR support, and facility management.

- **Business Question 2**

Which locations have the highest average employee salary?

- **Business Objective**

Calculate the average employee salary across office locations to compare compensation levels, support workforce planning, and assist management in evaluating salary distribution.

- **SQL Query**

```
SELECT  
  
    l.city, COALESCE(ROUND(AVG(e.salary), 2),0) AS avg_salary  
  
FROM  
  
    location l  
  
    LEFT JOIN  
  
    employee e ON l.location_id = e.location_id  
  
GROUP BY l.city , l.location_id  
  
ORDER BY avg_salary DESC;
```

- **Query Result**

City	AVG Salary
Ahmedabad	141433.77
Noida	140822.14
Hyderabad	140260.13
Delhi	138605.1
Pune	137564.21
Chennai	137248.3
Gurgaon	136810.53
Mumbai	136665.4
Bangalore	136546.12
Kolkata	134675.84

- **Business Insight**

Locations with the highest average salaries may represent regions with senior talent, specialized skills, or higher local labor costs. Comparing average salaries helps management understand compensation trends across different offices.

- **Business Question 3**

Which locations have the most experienced workforce on average?

- **Business Objective**

Calculate the average employee experience for each location to identify offices with the most experienced workforce and support workforce planning and talent management decisions.

- **SQL Query**

```
SELECT
    l.city, COALESCE(ROUND(AVG(e.experience_years), 2),0) AS
    avg_experience_years
FROM
    location l
    LEFT JOIN
    employee e ON l.location_id = e.location_id
GROUP BY l.city , l.location_id
ORDER BY avg_experience_years DESC;
```

- Query Result

City	AVG_Experience_Years
Kolkata	10.38
Gurgaon	10.32
Hyderabad	10.3
Delhi	10.3
Chennai	10.26
Mumbai	9.98
Bangalore	9.95
Pune	9.81
Noida	9.53
Ahmedabad	9.47

- Business Insight

Locations with higher average employee experience represent mature teams that may be well suited for leadership, mentoring, and managing complex business operations.
